

Analog Electronics — Syllabus

Unofficial Course Title: Electronics

*Spring 2026, Deep Springs College, **Prof. Brian Hill***

Course meeting times: 8:00-9:25am, Tu/Fr

Overview

To understand where this course fits into the electronics landscape, it is useful to crudely separate three eras of electronics history:

- There was a Frankenstein era going all the way back to the roots of the field where devices were expensive, dangerous, not very capable, and initially not very widespread. As vacuum tubes became mass produced in the 1920s, the Frankenstein era started to close.
- From 1947 onward, when the transistor was invented, and immediately began to obsolete the previous high-voltage vacuum tubes. Transistors and diodes in myriad variety were developed, all dependent on a new material known as a semiconductor. This ushered in a Goldilocks era, where it became far easier to build circuits.
- In the 1980s, the success of the semiconductor era ushered in the present era. The discrete transistor was almost completely replaced by miniature versions of itself replicated *en masse* in integrated circuits and silicon wafers, and even in the most mundane circuitry like that in toasters and thermostats, the componentry became hidden in epoxy-coated boxes that the consumer had no way and little reason to tinker with.

What is the point, today, of going back to the technology of the 1970s and tinkering with circuits containing only discrete components and some simple integrated circuits, when a general-purpose chip with billions of transistors is already in your pocket, and most anybody with a little computer knowledge can program complicated things with no electronics artistry at all? The point is to actually understand electronics, and be among the few empowered to do genuinely new things with electronics.

Prerequisite

The only prerequisite for this class is good high school mathematics. When I say “high school mathematics” I am specifically distinguishing it from “college mathematics” even though many students on a college prep track take AP calculus while still in high school. More formally, what I am referring to by “high school mathematics” is the four years consisting of: Algebra I; Geometry (plane geometry with proofs); Algebra II (with trigonometry); and Precalculus (logarithms, exponentials, polynomials, rational functions, and infinite sequences and series). Good high school mathematics will make the mathematical descriptions of analog circuitry readily intelligible.

If your high school mathematics is rusty, do not let that be a deterrent. It just means that you will have to work harder to polish it back up, and actually, this course is the perfect place to do that.

Materials

In this course, we will return to the Goldilocks era and work through the same materials that countless hobbyists in the 1970s used to learn electronics. This is an incredible sweet spot that is sadly getting lost. The materials of the Goldilocks era were hands-on, accessible, and remarkably educational.

Our course will cover the associated theory. The bulk of your time will be dominated by hands-on work with Radio Shack kits and Workbook 1 of the companion text authored by Forrest Mims. We will spend ten or eleven weeks of the semester building Mims’ introductory circuits, and then in the final three or four weeks, we’ll have time for each of you to choose and independently build a complete electronics project.

Unit Outline (rough and will probably have additions!)

- Unit 1: Introduction to the Lab, and to Basic Electronics Concepts
- Unit 2: Discrete Component Types and Inventory
- Unit 3: Circuit Diagrams, Circuit Assembly Techniques
- Unit 4: Troubleshooting
- Unit 5: Controlling Circuits with Switches and Relays
- Unit 6: Using Resistors and Potentiometers
- Unit 7: Capacitors and RC Circuits
- Unit 8: Transformers, Speakers, and Microphones
- Unit 9: Semiconductors I: Diodes and LEDs
- Unit 10: Semiconductors II: Transistors, Amplifiers, Voltage Regulators
- Unit 11: Special Project: FM receiver kit

Grading

- 60% lab assignments (demonstration of your circuitry in almost every class)
- 25% written problem sets (where you will practice and demonstrate your understanding of circuit diagrams and theory)
- 15% final project (the last 3 of our 14 weeks) — most of April — will be devoted to building an AM/FM receiver from a kit

Problem Sets / Handouts / Being Neat and Organized

There will be problem sets, problem set solutions, two exams and their solutions, and many handouts. To be organized, locate a three-ring binder, and file everything chronologically. Actually, reverse-chronological is the most convenient, because then you always open your binder to what you are currently working on.

Problem sets should be on standard 8 1/2 x 11 paper. Multi-page problem sets should be stapled. Corrections should be erased (if done in pencil) or recopied (if done in pen) if necessary to make a neat result. To make nice diagrams and graphs, you will very often need a ruler.

Absences and Late Work

The College's policies on absences and late work are applicable except that I don't require as much advance notice as is specified in the Deep Springs Handbook: specifically, one day's advance notice of an absence (or a need to turn in late work) is acceptable for our class. See the Handbook for additional details.